

Machine Learning ke 3 Types

Handwritten notes ko organise karke, examples ke saath detail mein samjhaya gaya

Machine Learning ko mainly **3 categories** mein baanta jata hai — is baat ko dhyaan mein rakhna bahut zaroori hai, kyunki isi se pata chalta hai ki kaunsa algorithm kab use karna hai:

Type	Data kaisa hota hai?	Kaam kya hota hai?
1. Supervised ML	Labeled data (Input + Output diya hota hai)	Prediction karna (Regression / Classification)
2. Unsupervised ML	Unlabeled data (Sirf Input, Output nahi diya)	Patterns / Groups dhoondna (Clustering)
3. Reinforcement Learning	Environment se interaction (Reward / Penalty)	Trial & error se best action seekhna

Neeche har type ko ek-ek karke, real-life example ke saath, step-by-step samjhaya gaya hai.

1 | Supervised Machine Learning

Supervised ML mein hume dataset already **labeled** milta hai — matlab har row mein **Input (Independent Features)** ke saath uska correct **Output (Dependent Feature)** bhi diya hota hai. Model in dono ko dekh kar seekhta hai, taaki naye input par sahi output predict kar sake.

Supervised ML ke andar 2 techniques aati hain:

- **Regression** — jab output **continuous** (number) ho, jaise price, temperature, salary.
- **Classification** — jab output **categorical** (fixed categories) ho, jaise Pass/Fail, Yes/No.

Yeh decide karne ka simple tareeka:

Step	Kya dekhna hai
①	Dataset mein Dependent (Output) feature kaunsa hai — usse identify karo
②	Agar wo output Continuous hai (koi bhi number ho sakta hai) → Regression
③	Agar wo output Categorical hai (fixed limited categories) → Classification

1.1 Regression — Example: House Price Prediction

Maan lo humein ghar ka **price predict** karna hai. Yahan **Size of House** aur **Number of Rooms** independent features hain (input), aur **Price** dependent feature hai (output) — jo predict karna hai.

Size of House (sq ft)	# of Rooms	Price (Output)
5000	5	■450K
6000	6	■500K
...	...	...

Yahan **Price** ek **continuous value** hai — yeh 450K, 500K, 450.5K ya kuch bhi ho sakta hai. Isliye yeh ek **Regression Problem** hai.

Regression ke common algorithms:

- ① **Linear Regression** — sabse basic, ek straight line fit karta hai data par
- ② **Ridge & Lasso Regression** — Linear Regression ka improved version (overfitting kam karta hai)
- ③ **ElasticNet** — Ridge aur Lasso dono ka combination

1.2 Classification — Example: Pass / Fail Prediction

Ab dekhte hain ek doosra example — student ke **Study Hours** aur **Play Hours** ke basis par yeh predict karna ki wo **Pass** karega ya **Fail**.

No. of Study Hours	No. of Play Hours	Result (Output)
7	3	Pass

No. of Study Hours	No. of Play Hours	Result (Output)
2	6	Fail

Yahan output sirf **2 fixed categories** mein hai — Pass ya Fail. Yeh koi random number nahi hai, isliye yeh **Classification Problem** hai.

Classification ke 2 types hote hain:

Type	Matlab	Example
Binary Classification	Sirf 2 possible outputs (categories)	Pass / Fail
Multiclass Classification	2 se zyada possible outputs (categories)	May Be / Pass / Fail

Classification ke common algorithms:

- ④ **Logistic Regression** — naam mein 'Regression' hai, lekin yeh actually Classification ke liye use hoti hai
- ⑤ **Decision Tree** — dono, Classification aur Regression, ke liye use hota hai
- ⑥ **Random Forest** — multiple Decision Trees ka combination (dono ke liye use hota hai)
- ⑦ **AdaBoost** — boosting algorithm (dono ke liye use hota hai)
- ⑧ **XGBoost** — powerful boosting algorithm, bahut popular (dono ke liye use hota hai)

Note: Decision Tree, Random Forest, AdaBoost aur XGBoost — yeh chaaron algorithms **Classification aur Regression**, dono problems solve kar sakte hain.

2 | Unsupervised Machine Learning

Unsupervised ML mein dataset mein sirf **input** hota hai, koi **output/label nahi diya hota**. Model khud hi data mein **similar patterns ya groups (clusters)** dhoondta hai.

Example: Customer Segmentation (E-commerce Company)

Ek E-commerce company apne customers ka data collect karti hai — **Salary** aur **Spending Score (1-10)**. Ab company yeh jaanna chahti hai ki customers ko kaise groups mein baanta ja sakta hai, taaki har group ko targeted marketing ki ja sake.

Salary	Spending Score (1-10)
20,000	9
45,000	2
...	...

Jab is data ko **Salary vs Spending** graph par plot karte hain, to Unsupervised algorithm khud hi customers ko alag-alag **clusters (groups)** mein baant deta hai — jaise:

- **Cluster 1 (Orange):** Low Salary, High Spending
- **Cluster 2 (Yellow):** High Salary, High Spending
- **Cluster 3 (Green):** High Salary, Low Spending

Is■■■ baad, company un customers ko target kar sakti hai jinki **spending kam hai** — unhe email par **discount offers** bhej kar wapas engage kiya ja sakta hai. Yehi Unsupervised ML ka real-world use hai — bina kisi output label ke, sirf data ke patterns se useful insights nikalna.

Unsupervised ML ke common algorithms:

- ① **K-Means Clustering** — data ko K number ke groups mein baantta hai
- ② **Hierarchical Clustering** — clusters ka ek tree/hierarchy banata hai
- ③ **DBSCAN Clustering** — density ke basis par clusters banata hai (outliers bhi identify karta hai)

3 | Reinforcement Learning

Reinforcement Learning (RL) mein koi labeled dataset nahi hota. Iski jagah, ek **Agent** apne **Environment** ke saath interact karta hai, actions leta hai, aur uske result mein **Reward** (achha action) ya **Penalty** (galat action) milta hai.

Time ke saath, Agent trial-and-error se seekhta hai ki kaunsa action **maximum reward** deta hai, aur wahi behavior improve karta rehta hai.

Concept	Matlab
Agent	Wo system jo decisions leta hai (jaise robot, game player)
Environment	Wo duniya jisme Agent kaam karta hai
Action	Agent ka har decision / step
Reward / Penalty	Feedback — achhe action par reward, galat par penalty

Real-life examples:

- Self-driving cars — road par sahi decisions lena seekhna
- Game-playing AI (jaise Chess, AlphaGo) — jeetne wali strategy seekhna
- Robotics — chalna, pakadna jaisi physical tasks seekhna

Quick Summary — Sab Kuch Ek Nazar Mein

Type	Data	Goal	Key Algorithms
Supervised (Regression)	Labeled (continuous output)	Number predict karna	Linear Reg, Ridge/Lasso, ElasticNet
Supervised (Classification)	Labeled (categorical output)	Category predict karna	Logistic Reg, Decision Tree, Random Forest, AdaBoost, XGBoost
Unsupervised	Unlabeled	Groups/patterns dhoondna	K-Means, Hierarchical, DBSCAN
Reinforcement Learning	Reward/Penalty (no dataset)	Best action seekhna	Q-Learning, Deep Q-Network, Policy Gradient

Yaad rakhne ka trick: Dependent/Output feature dikh raha ho → Supervised (continuous → Regression, categorical → Classification). Output feature na ho, sirf groups dhoondne ho → Unsupervised. Reward/Penalty ke through seekhna ho → Reinforcement Learning.

— Notes organised & explained —